

RAUC: A Novel Geometric Measure for Point Processes Applied to Machine Learning

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Abstract—We introduce the Relative Area Under the Curve (RAUC) measure as a geometric criterion for analyzing the structure of point processes. It is applicable in various machine learning methods, including the analysis of predictors in regression models and the study of the plane resulting from dimensionality reduction techniques, such as the plane defined by the first two principal components in Principal Component Analysis (PCA) or the two dimensional projection obtained through non-linear methods like Uniform Manifold Approximation and Projection (UMAP).

The variables defining the bidimensional representation are normalized using a min-max transformation to ensure they lie within the unit square. The proposed measure compares the empirical empty-space loss function, derived from the alpha-shape of the point process, with a reference curve obtained under a complete spatial randomness (CSR) point process, defining RAUC as the ratio between the area under the CSR curve and the area under the empirical loss function. This provides a scalar measure of deviation from spatial randomness, where RAUC values close to 0 indicate strong clustering patterns, and values moving away from 0 begin to reflect patterns approaching randomness.

We evaluate the proposed method on simulated and classical machine learning datasets, showing that RAUC provides a useful and interpretable measure that can be integrated both as a pre-processing tool and as a complementary criterion for analyzing the results of popular machine learning techniques.

Index Terms—Empty-Space Loss Function, Point Processes, Geometric Patterns, Spatial Randomness, Clustering, Machine Learning.

I. INTRODUCTION

The analysis of point processes is central to many tasks in machine learning, where data are commonly represented as point clouds in spaces of varying dimension, but are often examined through bidimensional representations. In regression analysis, bivariate relationships between predictors and the response variable are routinely explored to identify patterns of dependence, including linear and nonlinear structures, as well as associations between variables [2]. Similarly, dimensionality reduction techniques, such as the plane defined by the first two principal components in PCA or projections obtained through non-linear methods like UMAP, enable the study of the geometric structure of data in two dimensions to reveal clusters and latent patterns.

It is important to note that dependence between variables is not limited to linear relationships. In many cases, more complex nonlinear relationships may arise, inducing different geometric structures in the plane. In this context, classical measures such as Pearson correlation are limited, as they only

quantify linear association between variables. More general dependence measures, while capable of detecting nonlinear associations, often lack a direct geometric interpretation of the underlying structure [4].

On the other hand, in the context of point processes, randomness typically indicates weak or no dependence between variables, as no clear linear or nonlinear trend is observed [2]. Consequently, the analysis of bidimensional point patterns becomes essential to distinguish between configurations close to spatial randomness and those exhibiting clustering or regularity, reflecting different types of dependence between variables.

In this work, the Relative Area Under the Curve (RAUC) measure is proposed as a geometric criterion for analyzing the structure of bidimensional point processes, providing a quantitative measure of the degree of randomness or clustering exhibited by a point cloud. This, in turn, enables a principled interpretation of bivariate relationships that arise at different stages of machine learning techniques.

The approach is based on representing each pair of variables $X, Y \in \mathbb{R}^n$ as a point cloud in the plane, from which geometric objects based on alpha-shapes are constructed. From this construction, an empty-space loss function is defined, allowing comparison between the observed structure and the CSR point process.

In this study, the RAUC measure is incorporated into machine learning techniques, considered in two stages: prior to the construction of regression models and during the analysis of results from dimensionality reduction techniques. Experiments are conducted on both simulated data and classical machine learning datasets. This aims to highlight the use of RAUC in machine learning applications, positioning it as a complementary and interpretable geometric tool alongside widely used classical measures.

The main contributions of this work are summarized as follows:

- We introduce RAUC as a new geometric measure for analyzing the structure of bidimensional point processes, identifying patterns that range from clustering to complete spatial randomness (CSR).
- We demonstrate the integration of RAUC into different stages of machine learning workflows, performing variable selection based on geometric structures with the target variable prior to the construction of regression

models, and analyzing bidimensional projections of high-dimensional data.

- We validate the proposed approach through experiments on both simulated data and classical machine learning datasets, illustrating its effectiveness and versatility.

II. THEORETICAL FRAMEWORK

A. Distance-Based and Empty-Space Characterizations of Point Processes

1) *Distances in point patterns*: When analyzing point pattern data, the goal is to determine whether the points are distributed independently or if they exhibit some form of geometric dependence. Figure 1 shows the three principal patterns; regularity, independence, and clustering.

To quantify this dependence, correlation is commonly used. However, correlation is merely a summary measure of statistical association and does not provide a characterization of causality. Furthermore, classical correlation measures are limited to capturing linear relationships and may fail to detect complex nonlinear patterns and do not provide information about the geometric structure of the point patterns, which can be crucial for understanding the nature of the dependence between variables [3].

To enrich the analysis of point patterns, it is useful to examine the spatial arrangement of points by measuring interpoint spacings and minimum distances [2]. Therefore, a comprehensive analysis of point patterns should consider both aspects. Within this approach, various types of distances can be quantified, including the following

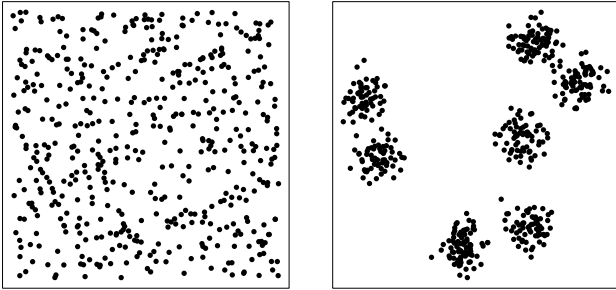


Fig. 1. Bidimensional point patterns of independence (left) and clustering (right).

- **Pairwise distance**: $d_{ij} = \|x_i - x_j\|$ between all pairs of points x_i and x_j with $i \neq j$ in the pattern.
- **Nearest-neighbour distance**: $d_i = \min_{j \neq i} d_{ij}$, the distance from a data point x_i to the nearest other data point.
- **Empty-space distance**: $d(u) = \min_j \|u - x_j\|$, the distance from a fixed location u to the nearest data point [4].

2) *Empty Space Function for a Point Process*: If X is a spatial point process, the empty space distance at a location $u \in \mathbb{R}^2$ is defined as $d(u, X) = \min\{\|u - x_i\| : x_i \in X\}$, that is, the distance from u to the nearest point of the process X .

A point process is considered stationary if, when observed through a window W , its statistical properties are invariant

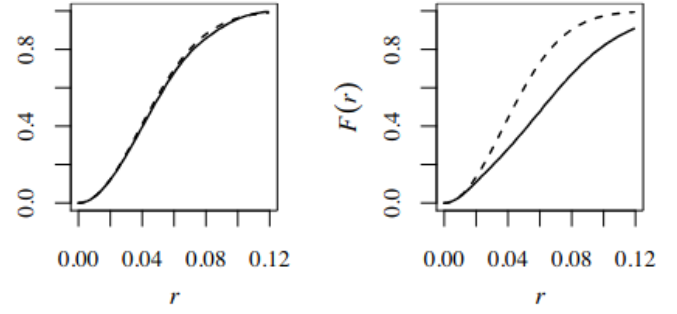


Fig. 2. Empirical empty-space function F (solid lines) for each pattern in Figure 1, and the theoretical function for a Poisson process (dashed lines).

Note. Reprinted from *Spatial Point Patterns: Methodology and Applications with R*, by A. Baddeley, E. Rubak, and R. Turner, 2015, CRC Press.

with respect to the location of that window in bidimensional space [2].

In a stationary point process, the cumulative distribution function of the empty-space distance is defined as $F(r) = P\{d(u, X) \leq r\}$, where $r \geq 0$ and u is an arbitrary location. The function $F(r)$ is non-decreasing, with $F(0) = 0$, and increases as r grows [2].

3) *Intensity*: The intensity λ is a descriptive measure of a point process and is interpreted as the expected number of points per unit area.

A point process X is said to have homogeneous intensity if, for any bounded region $W \subset \mathbb{R}^2$, the expected number of points in W , denoted $n(X \cap W)$, is $E[n(X \cap W)] = \lambda|W|$, where λ is a constant called the intensity [2].

To estimate the intensity from observed data, the empirical point density is used $\hat{\lambda} = \frac{n(X)}{|W|}$, where $|W|$ is the area of the observation window, and $n(X)$ denotes the total number of observed points.

In particular, if the point process follows a homogeneous Poisson model, the total number of points satisfies $n(X \cap W) \sim \text{Poisson}(\lambda|W|)$, with mean $\lambda|W|$ and variance $\frac{\lambda}{|W|}$.

4) *Empty Space Function for a Complete Spatial Randomness Process*: If X is a CSR process in $\mathbb{R}^2$ with intensity λ , the number of points falling in a set W follows a Poisson distribution with mean $\lambda|W|$. The theoretical empty-space function for CSR is defined as $P(r) = 1 - e^{-\lambda\pi r^2}$ [2]. In this type of spatial randomness processes, the mean is $E[d(u, X)] = \frac{1}{2\sqrt{\lambda}}$ and the variance is $\text{Var}[d(u, X)] = \frac{4-\pi}{4\pi\lambda}$.

5) *Interpretation of the Empty Space Function*: Figure 2 shows the estimated empty-space function for the two patterns presented in Figure 1. In the left panel, the empirical and theoretical curves closely coincide, indicating compatibility with complete spatial randomness process. Conversely, in the right panel, the empirical curve lies below the theoretical curve, indicating a clustered pattern.

B. Geometric goodness-of-fit index on classical data

The geometric goodness-of-fit index proposed by Hernández and Solís [5] provides a measure that, for a fixed radius α of the alpha-shape of the bidimensional projection of two variables $X = (x_1, \dots, x_n)^t, Y = (y_1, \dots, y_n)^t \in \mathbb{R}^n$, evaluates the geometric correlation between these variables.

Let $\alpha > 0$ be the radius parameter of the alpha-shape $\mathcal{R}_\alpha$ of the bidimensional projection (X, Y) , computed using the Delaunay triangulation procedure described in [4]. The area of the alpha-shape is denoted by $\text{Area}(\mathcal{R}_\alpha)$, and the area of the smallest rectangle enclosing the bidimensional projection is denoted by W and is calculated by

$$\text{Area}(W) = \left(\max_{1 \leq i \leq n} \{x_i\} - \min_{1 \leq i \leq n} \{x_i\} \right) \left(\max_{1 \leq i \leq n} \{y_i\} - \min_{1 \leq i \leq n} \{y_i\} \right)$$

For each α , the geometric goodness-of-fit index is defined as

$$R_{\text{Geom}, \alpha}^2 = 1 - \frac{\text{Area}(\mathcal{R}_\alpha)}{\text{Area}(W)}.$$

This quantity measures the proportion of empty space remaining inside W once the alpha-shape is inserted. By construction, $0 \leq R_{\text{Geom}, \alpha}^2 \leq 1$.

For a fixed α , values close to 1 indicate that the point cloud occupies a small portion of the bounding region and therefore displays a geometric structure. Values near 0 suggest that the configuration is more similar to CSR process.

To study the behavior across scales, the authors defined the mapping

$$f: \mathbb{R}^+ \rightarrow [0, 1], \quad f(\alpha) = R_{\text{Geom}, \alpha}^2.$$

A full characterization of the geometric structure of the bidimensional projection requires analyzing the entire curve f . The evolution of this curve reflects how the configuration progressively fills the domain as the radius increases.

C. Machine Learning

Machine learning analyzes patterns in data to understand underlying processes, make predictions, and uncover structure [11]. It is widely applied across diverse domains, including finance, manufacturing, and healthcare.

Two classical machine learning methods are regression and dimensionality reduction, which are described below.

1) *Regression Methods*: Regression models are widely used in statistical learning to predict a continuous outcome variable based on one or more explanatory variables. Traditional approaches, such as linear regression, assume a linear relationship between the predictors and the response, estimating coefficients that minimize the residual sum of squares. Extensions include generalized linear models and regularized regression methods that help control overfitting and improve predictive performance, particularly in high-dimensional settings [11].

Beyond linear models, a variety of machine learning approaches can be employed for regression tasks. These include

methods like Random Forests, Support Vector Machine and Neural Networks, among others. According to [11] these models are capable of capturing non-linear relationships between predictors and the outcome, and often achieve higher predictive accuracy when sufficient data are available.

To assess the performance and generalization of regression models, k -fold cross-validation is commonly used. In this technique the dataset is partitioned into k subsets; the model is trained on $k - 1$ folds and tested on the remaining fold [11]. This process is repeated k times, and performance metrics such as Mean Absolute Error (MAE) or Root Mean Square Error (RMSE) are averaged across folds, providing a robust estimate of the predictive ability of the model on unseen data.

2) *Dimensionality Reduction Techniques*: Dimensionality reduction techniques aim to simplify high-dimensional data by projecting it onto a lower-dimensional space while preserving its essential structure.

- *Principal Component Analysis (PCA)*:

Principal Component Analysis (PCA) reduces the dimensionality of high-dimensional data while preserving most of its variability [10].

Let $X \in \mathbb{R}^{n \times p}$ be a dataset with n observations and p variables, and let $x_i \in \mathbb{R}^p$ denote the i -th observation. PCA seeks a reduced set of derived variables (components) that retain most of the information in X . The first principal component for observation x_i is $z_{i1} = \alpha_1^\top x_i$, where $\alpha_1 \in \mathbb{R}^p$ is chosen to maximize $\text{Var}(z_{i1})$ subject to $\|\alpha_1\| = 1$. This procedure is repeated for $k = 2, \dots, p$, with each component $z_{ik} = \alpha_k^\top x_i$ being uncorrelated with all preceding components while capturing the largest remaining variance.

If X has covariance matrix $\Sigma \in \mathbb{R}^{p \times p}$ with entries $\sigma_{ij} = \text{Cov}(X_i, X_j)$, then each α_k is the eigenvector of Σ corresponding to the k -th largest eigenvalue λ_k , such that $\text{Var}(z_{ik}) = \lambda_k$ [10].

- *Uniform Manifold Approximation and Projection (UMAP)*:

UMAP constructs a low-dimensional representation of high-dimensional data while preserving its topological and metric structure [6].

Let $X = \{x_1, \dots, x_n\} \subset \mathbb{R}^p$ be a dataset assumed to lie on a Riemannian manifold M with metric g . UMAP approximates local neighborhoods by computing distances from each point x_i to its k -nearest neighbors and normalizes these distances to account for local density variations, producing pseudo-metric spaces $\{(x_i, d_i)\}_{i=1}^N$.

Each local space is transformed into a fuzzy simplicial set, which are then merged into a global fuzzy simplicial set to capture the overall manifold structure. A low-dimensional embedding $y_i \in \mathbb{R}^d$ is obtained by optimizing the layout to minimize the discrepancy between high-dimensional and low-dimensional representations using stochastic gradient descent [6].

III. METHODOLOGY

A. RAUC for classical point-valued data

Consider a classical numerical dataset represented by a matrix $X \in \mathbb{R}^{n \times (m+1)}$, where each row corresponds to an observation and each column $X^j = (x_{1j}, \dots, x_{nj})^\top \in \mathbb{R}^n$, for $j = 1, \dots, m$, represents a numerical variable. The remaining column $Y = X^{m+1} = (x_{1,m+1}, \dots, x_{n,m+1})^\top = (y_1, \dots, y_n)^\top \in \mathbb{R}^n$ is the target variable with respect to which the geometric structure of each predictor is evaluated in the point clouds.

Thus, for each X^j , $j = 1, \dots, m$, the relationship with the variable Y can be represented through a scatter plot

$$N_{XY}^j = \{(x_{ij}, y_i) : i = 1, \dots, n\},$$

which allows the study of the geometric structure of the resulting point cloud.

In order to remove scale and location effects and ensure comparability between variables, a min-max normalization should be applied to each column of the matrix X . This transformation maps all variables into the unit interval $[0, 1]$. Consequently, for each variable $j = 1, \dots, m$ and for the target variable Y , the corresponding normalized vectors are defined as $\tilde{X}^j = (\tilde{x}_{1j}, \dots, \tilde{x}_{nj})^t \in \mathbb{R}^n$ and $\tilde{Y} = (\tilde{y}_1, \dots, \tilde{y}_n)^t \in \mathbb{R}^n$.

The min-max transformation is given by

$$\begin{aligned} \tilde{x}_{ij} &= \frac{x_{ij} - \min_{1 \leq i \leq n} \{x_{ij}\}}{\max_{1 \leq i \leq n} \{x_{ij}\} - \min_{1 \leq i \leq n} \{x_{ij}\}}, \\ \tilde{y}_i &= \frac{y_i - \min_{1 \leq i \leq n} \{y_i\}}{\max_{1 \leq i \leq n} \{y_i\} - \min_{1 \leq i \leq n} \{y_i\}}. \end{aligned}$$

This induces a normalized point cloud

$$N_{\tilde{X}\tilde{Y}}^j = \{(\tilde{x}_{ij}, \tilde{y}_i) : i = 1, \dots, n\} \subset W = [0, 1]^2,$$

which represents the joint geometric structure between the selected predictor and the response variable in a rescaled coordinate system. This transformation ensures that the subsequent geometric analysis depends exclusively on the spatial configuration of the points, rather than on the original measurement units or the scale of the variables.

The geometry of each point cloud $N_{\tilde{X}\tilde{Y}}^j$ is captured by its associated alpha-shape $\mathcal{R}_\alpha^j$, which provides a continuous geometric approximation of the scatter plot. The geometric goodness-of-fit of the point cloud within the unit square $W = [0, 1]^2$ is quantified by the index

$$I_\alpha^j = 1 - |\mathcal{R}_\alpha^j|,$$

where $|\mathcal{R}_\alpha^j|$ denotes the area of the alpha-shape for a fixed value of α .

The associated empty-space loss function is defined, for each predictor j , as

$$f^j : \mathbb{R}^+ \rightarrow [0, 1], \quad f^j(\alpha) = I_\alpha^j.$$

In practice, the function f^j is evaluated on a discrete grid $\{\alpha_0, \alpha_1, \dots, \alpha_n\}$, with $\alpha_0 = 0$, yielding a stepwise (piecewise-constant) representation defined by

$$f^j(\alpha) = \begin{cases} f^j(\alpha_1), & \alpha \in [\alpha_0, \alpha_1), \\ f^j(\alpha_2), & \alpha \in [\alpha_1, \alpha_2), \\ \vdots & \\ f^j(\alpha_n), & \alpha \in [\alpha_{n-1}, \alpha_n]. \end{cases}$$

Under the null model of CSR, the point pattern is modeled as a homogeneous Poisson process with intensity $\lambda = n$ over W , and the corresponding theoretical empty-space function is given by

$$P_{\tilde{X}\tilde{Y}}(\alpha) = e^{-\lambda\pi\alpha^2}.$$

Finally, the Relative Area Under the Curve (RAUC) measure is defined, for each variable $j = 1, \dots, m$, as the ratio between the area under the CSR curve and the area under the empirical empty-space loss function:

$$\text{RAUC}^j = \frac{\int_0^{\alpha_n} e^{-\lambda\pi\alpha^2} d\alpha}{\sum_{i=0}^{n-1} f^j(\alpha_{i+1})(\alpha_{i+1} - \alpha_i)}.$$

This formulation provides a scale comparison between observed point clouds and the CSR benchmark, while preserving the discrete evaluation induced by the alpha-shape construction, offering a criterion for geometric variable relevance assessment in classical data.

B. Interpretation of the RAUC measure

The Relative Area Under the Curve (RAUC) measure provides a scalar summary of the geometric structure of each point process. It is defined as the ratio between the area under the reference curve for complete spatial randomness (CSR) and the area under the empirical empty-space loss function $f^j(\alpha)$. Since both quantities are strictly positive, $\text{RAUC} \in (0, \infty)$.

Values of RAUC close to 0 indicate strong clustering patterns, where points occupy a small portion of the unit square and empty space is abundant. As RAUC increases, moving away from 0, the observed patterns increasingly resemble CSR, reflecting weaker geometric dependence between variables or more uniform spatial distributions. Given that all data are normalized to the unit square $[0, 1] \times [0, 1]$ via a min-max transformation, RAUC values are typically moderate, yet remain positive, providing a consistent scale for interpretation.

RAUC offers an interpretable and versatile criterion for analyzing bivariate relationships in point processes at different stages of machine learning workflows. It can be used to rank predictor variables prior to the construction of regression models based on their geometric association with the response variable. Additionally, it is applicable for evaluating the geometric structure of bidimensional projections obtained through dimensionality reduction techniques such as PCA or UMAP, supporting a principled analysis of clustering, randomness, and spatial regularity in both predictive and exploratory tasks.

IV. APPLICATIONS

A. Regression.

1) *Experiment 1: Simulation Dataset:* In this study, a simulated dataset was constructed to evaluate the proposed methodology under different types of dependency structures between predictors and the response variable. A total of $n = 600$ observations were generated, where the response variable Y was sampled from a uniform distribution $Y \sim \mathcal{U}(-2, 2)$.

Six predictors were then constructed to represent distinct geometric and relationships with Y . The first predictor, X_1 , was generated from a sinusoidal transformation of Y combining three frequencies and adding Gaussian noise ε_1 . The second predictor, X_2 , follows a quadratic relationship with Y , also with Gaussian noise, forming a parabolic nonlinear pattern.

The third predictor, X_3 , combines a linear trend with a high-frequency sinusoidal component, producing a mixed linear–nonlinear structure. The fourth predictor, X_4 , is generated as a simple linear function of Y with Gaussian noise, representing a classical linear dependence setting.

In contrast, X_5 was generated independently of Y from a uniform distribution over the same range, serving as a scenario with no apparent dependence structure. Finally, X_6 is constructed as a clustered process by dividing Y into three groups. Within each group, values are sampled from Gaussian distributions with distinct means, inducing a structure with clustering behavior.

This design allows for the evaluation of the proposed method across a various scenarios. The mathematical formulation of the described variables can be found in Table I.

TABLE I
DESCRIPTION OF THE SIMULATED PREDICTORS AND THE MECHANISMS USED TO GENERATE THEM.

Variable	Generating process
Y	$Y \sim \mathcal{U}(-2, 2)$
X_1	$\sin(Y) + 0.5 \sin(2Y) + \varepsilon_1, \varepsilon_1 \sim \mathcal{N}(0, 0.05)$
X_2	$Y^2 + \varepsilon_2, \varepsilon_2 \sim \mathcal{N}(0, 0.15)$
X_3	$Y + 0.8 \sin(3Y) + \varepsilon_3, \varepsilon_3 \sim \mathcal{N}(0, 0.1)$
X_4	$Y + \varepsilon_4, \varepsilon_4 \sim \mathcal{N}(0, 0.08)$
X_5	$X_5 \sim \mathcal{U}(\min(Y), \max(Y))$
X_6	$X_6 \sim \begin{cases} \mathcal{N}(-1.5, 0.2), & Y \in C_1 \\ \mathcal{N}(0, 0.2), & Y \in C_2 \\ \mathcal{N}(1.5, 0.2), & Y \in C_3 \end{cases}$

Once the RAUC ranking was obtained, a 5-fold cross-validation procedure was applied to identify the best subset of models with k predictors, for $k = 1, \dots, 6$, using a Random Forest model as the base estimator. For the model with a single predictor, the best performance was achieved using X_4 , obtaining an RMSE of 0.086. For $k = 2$, the best combination was X_1 and X_4 , with an RMSE of 0.071. For $k = 3$, the optimal subset was X_1, X_2 , and X_4 , achieving an RMSE of 0.059. For $k = 4$, the best model included X_2, X_3, X_4 , and X_6 , with an RMSE of 0.056. Finally, for $k = 5$, the selected variables were X_1, X_2, X_3, X_4 , and X_6 , resulting in an RMSE of 0.054.

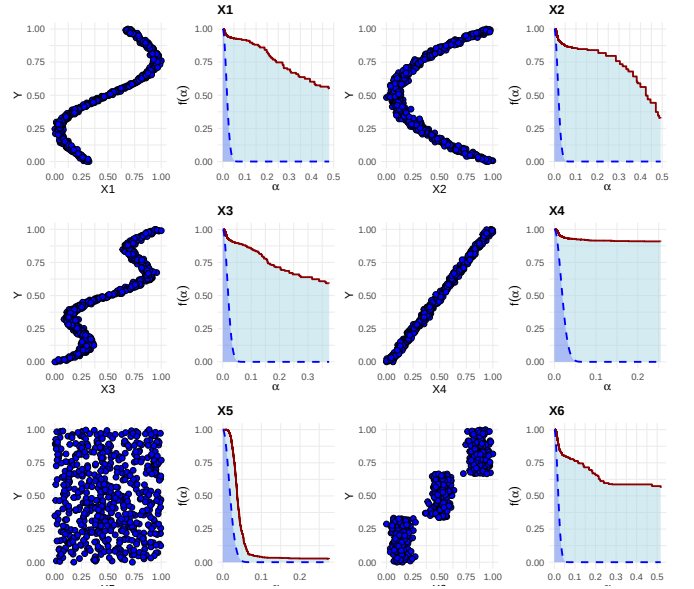


Fig. 3. Scatter plots of each predictor $X_i, i = 1, \dots, 6$, against the response variable Y are shown on the left, while the associated empty-space loss functions for each point process and the corresponding empty space loss function of the CSR process are displayed on the right.

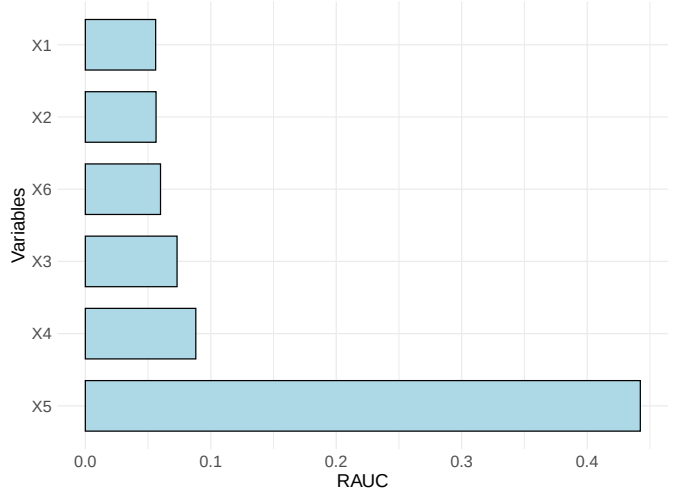


Fig. 4. RAUC ranking of the predictor variables in the simulated dataset. Lower RAUC values indicate a more persistent geometric structure in the predictor–response projection, whereas higher values suggest behavior closer CSR point processes defined on the unit square.

It is worth noting that the full model including all variables achieves the lowest overall error (RMSE = 0.043), and the use of Random Forest allows capturing nonlinear relationships without loss of generality.

2) *Experiment 2: Boston Dataset.:* In this experiment, the *Boston Housing* dataset available in the MASS package in R was used. This dataset contains 506 observations and 11 variables. The aim of the experiment is to rank the predictor variables according to their geometric relevance (RAUC) for modeling the regression relationship with the response variable, using a *Random Forest* model to capture potentially

nonlinear relationships without loss of generality. A brief description of the variables is provided below.

TABLE II
DESCRIPTION OF THE VARIABLES IN THE BOSTON HOUSING DATASET

Variable	Type	Description
crim	$\mathbb{R}$	Per capita crime rate by town.
zn	$\mathbb{R}$	Proportion of residential land zoned for lots over 25,000 square feet.
indus	$\mathbb{R}$	Proportion of non-retail business acres per town.
nox	$\mathbb{R}$	Nitric oxides concentration (parts per 10 million).
rm	$\mathbb{R}$	Average number of rooms per dwelling.
age	$\mathbb{R}$	Proportion of owner-occupied units built prior to 1940.
dis	$\mathbb{Z}$	Weighted mean of distances to five Boston employment centers.
tax	$\mathbb{R}$	Property tax rate per \$10,000.
ptratio	$\mathbb{R}$	Pupil-teacher ratio by town.
lstat	$\mathbb{R}$	Percentage of lower socioeconomic status population.
medv	$\mathbb{R}$	Median value of owner-occupied homes (in thousands of dollars).

In this experiment, `medv` is considered as the response variable, while the remaining variables are treated as predictors. In Figure 5, the scatter plots between each predictor and the target variable are displayed. Nonlinear patterns can be observed in most cases, along with some instances of approximately linear relationships with noise.

On the right-hand side of each point process, the corresponding empty-space loss function f is shown. These curves illustrate how the alpha-shape progressively fills the empty space within the unit square as α increases. Notably, variables such as `crim`, `rm` and `lstat` stabilize more rapidly than others, indicating that their associated point clouds are more compact and follow a persistent geometric structure. Consequently, the filling process is smoother and exhibits fewer abrupt changes compared to variables such as `age`, where the spatial distribution is more dispersed.

In Figure 6, the RAUC values are presented in ascending order. According to our measure, smaller values are associated with predictor variables that exhibit stronger geometric structure in their relationship with `medv`. In this sense, the variables `crim`, `rm` and `lstat` stand out as those with the most pronounced geometric patterns. This is consistent with their corresponding bidimensional scatter plots, which display clear and non-random structures, reflecting a dependence with the response variable.

From a practical perspective, this behavior is expected: higher crime rates tend to decrease housing attractiveness and prices, while the average number of rooms per dwelling is a key factor in property valuation.

In contrast, the variable `age` appears to be the most random, showing a weak geometric structure in its relationship with `medv`. This suggests that the proportion of owner-occupied units built prior to 1940 is less influential in determining the median housing value.

A total of $\binom{10}{3} = 120$ regression models, each including a combination of three predictor variables, were evaluated. The models were sorted according to the RAUC values of their predictors, from lowest to highest, to prioritize variables with stronger geometric structure. Among these, the combination `crim`, `rm`, and `lstat` achieved an RMSE of 0.087 under 5-fold cross-validation, placing it 4th in the overall ranking. The best-performing model included `rm`, `lstat`, and `nox` (ranked

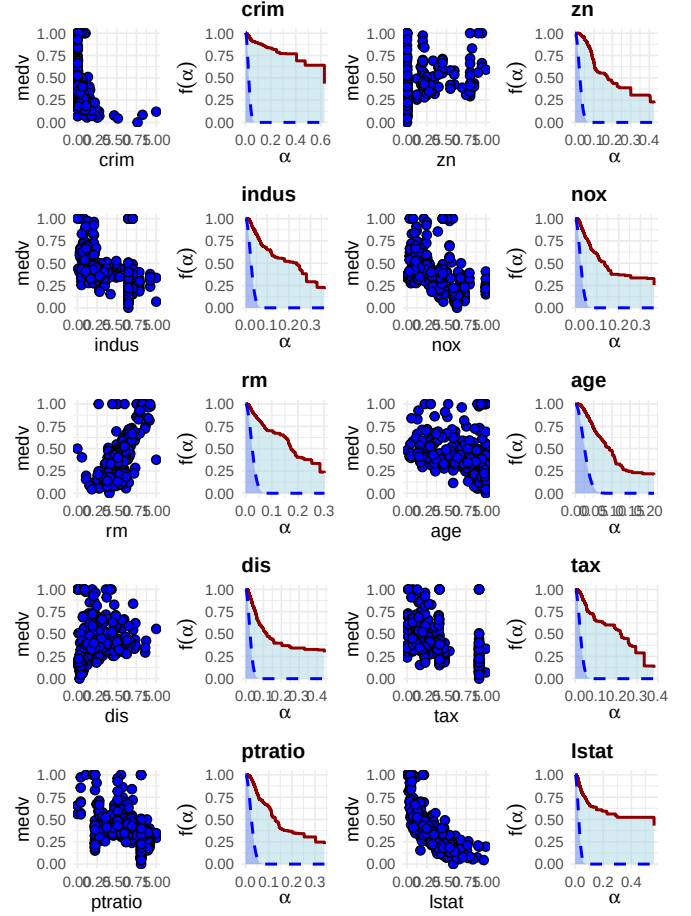


Fig. 5. Scatter plots of each variable versus the response variable `medv`, showing on the left the scatter plot of the points and on the right the associated empty-space loss function for each point process (solid lines), along with the corresponding empty-space loss function of the CSR process (dashed lines) on the unit square.

2nd, 3rd, and 8th in the RAUC ranking), obtaining an RMSE of 0.077.

Notably, the worst ten combinations all include the variable `age`, which is consistent with its near-random behavior. Additionally, among the top eight models with the lowest RMSE, the variables `lstat` consistently appear together. Overall, the models exhibit an RMSE range of $[0.077, 0.148]$.

The experiment was repeated using combinations of five variables, resulting in a total of $\binom{10}{5} = 252$ models. In this case, the best-performing model included the variables `rm`, `lstat`, `dis`, `tax`, and `ptratio`, achieving an RMSE of 0.077, with three of these variables belonging to the top five in the RAUC ranking. Interestingly, the model formed by the five highest-ranked variables according to RAUC was positioned 49th, with an RMSE of 0.088. Once again, the worst-performing models consistently included the variable `age`. The RMSE range remained within $[0.077, 0.148]$.

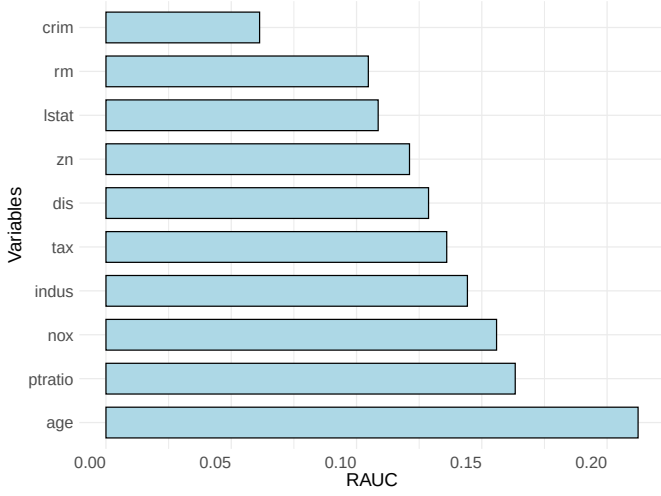


Fig. 6. RAUC ranking of the predictor variables in the Boston Housing dataset. Lower values of RAUC indicate a persistent geometric structure in the predictor-response projection, whereas higher values suggest a behavior closer to Complete Spatial Randomness (CSR) point processes.

B. Reduction of dimensionality

1) *Experiment 3: Comparison of the geometry of 2D UMAP projections under different hyperparameter settings for the Digit Recognizer dataset:* The dataset consists of grayscale images of handwritten digits, ranging from 0 to 9 obtained from Kaggle [1]. Each image is 28×28 pixels, resulting in 784 pixels per image. Each pixel is represented by an integer value between 0 and 255, where higher values indicate darker shades. For this experiment, a random sample of 785 observations was used.

The set contains 785 columns: the first column, called label, represents the digit in the image, and the remaining 784 columns store the pixel values, named `pixel0`, `pixel1`, ..., `pixel783`. Each pixel can be mapped to the 28×28 image matrix using the formula $x = i \cdot 28 + j$, where $i = 0, \dots, 27$ is the row index and $j = 0, \dots, 27$ is the column index.

We evaluated nine hyperparameter combinations for UMAP embeddings, varying the distance metric (Euclidean, Manhattan and Cosine) and the number of neighbors (5, 15 and 50). For each combination, the first two components of the resulting embedding were extracted to create a 2D projection, and the RAUC measure was computed to quantify the geometric structure of the point cloud.

The RAUC values were ranked from lowest to highest to identify the hyperparameter combination that produces the most structured and well-defined clusters. This ranking provides an criterion to compare hyperparameter settings based on the spatial organization of points in the 2D projection, beyond mere visual inspection.

As shown in Figures 7 and 8, the Cosine distance with 5 neighbors combination generates the most geometric structure, consistent with its graphical representation. In this projection, the point cloud exhibits the largest empty space and the most compact and well-formed clusters. In contrast, the Manhattan

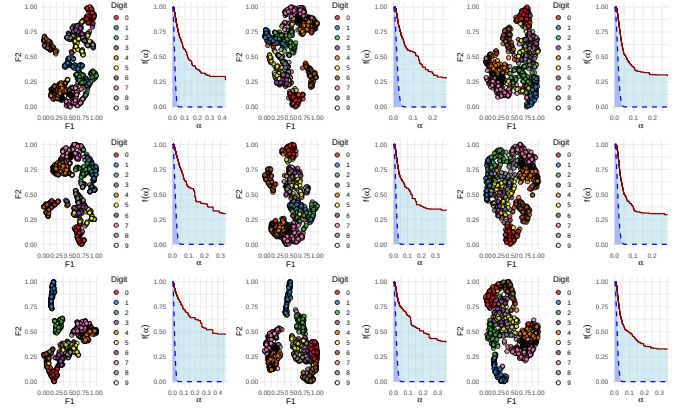


Fig. 7. Bidimensional embedding of the first two UMAP components of the Digit Recognizer dataset. Rows correspond to Euclidean, Manhattan, and Cosine distances, respectively, and columns correspond to $n_{\text{neighbors}} = 5, 15$, and 50. All embeddings are defined on the unit square.

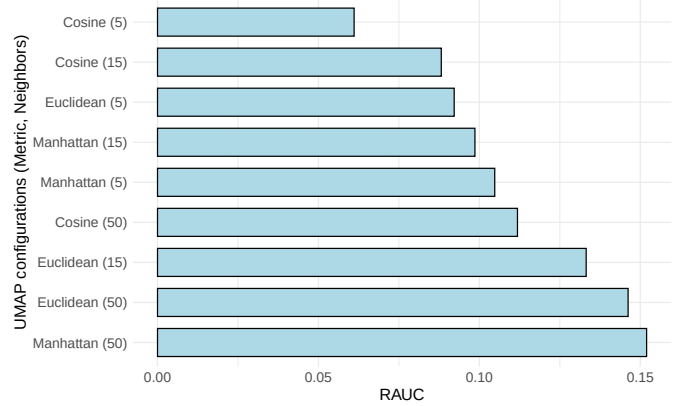


Fig. 8. RAUC ranking for different UMAP configurations. Lower values indicate persistent geometric structure in the bidimensional embedding, while higher values suggest behavior closer to CSR process on the unit square.

distance with 50 neighbors combination results in the least structured projection, where clusters are poorly separated and the points display a more random spatial arrangement.

2) *Experiment 4: Comparison of the geometry of 2D UMAP projections under different hyperparameter settings for the iris dataset:* In this experiment, the Iris dataset available in R was used, containing 150 observations corresponding to three flower species: *setosa*, *versicolor*, and *virginica*. Each observation includes four numerical features: `Sepal.Length`, `Sepal.Width`, `Petal.Length`, and `Petal.Width` and a categorical variable, `Species`, which indicates the species. The numerical features were used as input for the UMAP embeddings.

The same combinations of hyperparameters for distance metric and number of neighbors as in the previous experiment were tested. We observed that the optimal configuration corresponds to 50 neighbors with euclidean distance, achieving a RAUC value of 0.0719. This configuration yields the strongest geometric structure, as evidenced by the point pattern, which

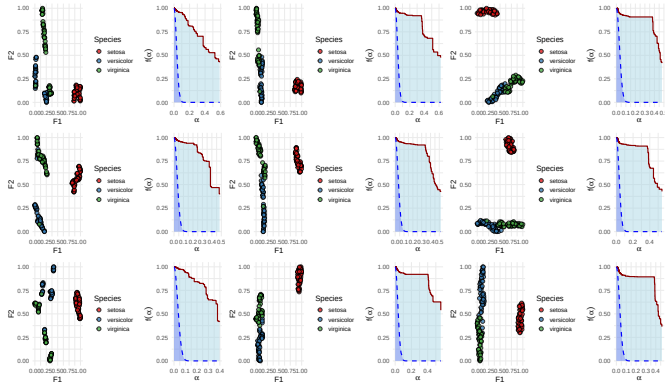


Fig. 9. Bidimensional embedding of the first two components. Rows correspond to euclidean, manhattan, and cosine distances, respectively, and columns correspond to $n_{\text{neighbors}} = 5, 15$, and 50 . All embeddings are defined on the unit square.

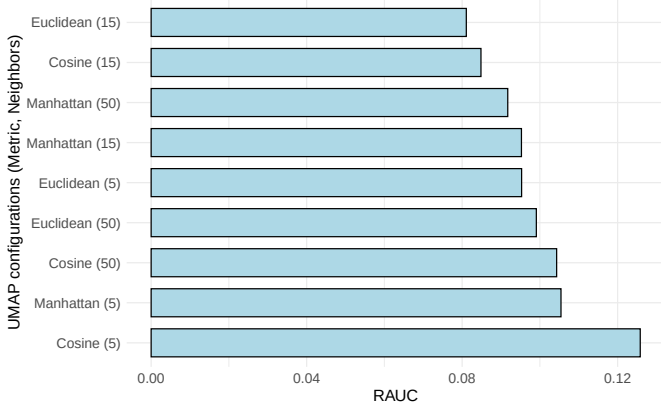


Fig. 10. RAUC ranking for different UMAP configurations. Lower values indicate persistent geometric structure in the bidimensional embedding, while higher values suggest behavior closer to CSR process on the unit square.

exhibits both high density and well-defined space organization.

A similar behavior is observed for cosine distance with 50 neighbors, which attains a RAUC value of 0.0829, making it the second-best configuration in terms of geometric preservation in the main embedding space. In contrast, the worst performing configuration is euclidean distance with 5 neighbors, where the points appear more dispersed and less structured in the projected space.

Regarding the empty space loss function, the best configuration shows a rapid stabilization of the curve at relatively large values of α , indicating that the alpha-shape remains largely stable as α increases. This suggests that the geometry is preserved across scales. In contrast, for the worst configuration, the alpha-shape changes rapidly as α increases, and the empty space loss decreases quickly before stabilizing at a lower value, reflecting weaker geometric persistence and less stable spatial structure.

3) *Experiment 5: Geometric comparison between PCA and UMAP bidimensional projections:* In this experiment, the RAUC index of the bidimensional projection obtained via

Principal Component Analysis (PCA) on the `iris` dataset was compared with that obtained using UMAP, in order to analyze the formation of clusters and geometric patterns.

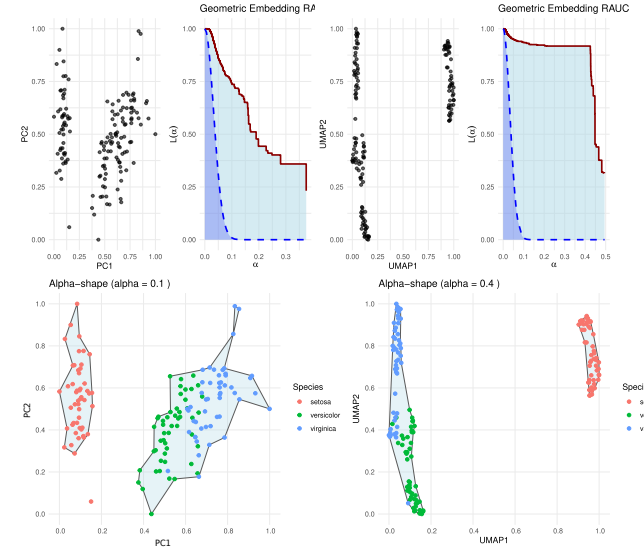


Fig. 11. Top: Comparison of the bidimensional projections obtained via PCA and UMAP, showing the empty-space loss functions and the CSR point process in the unit square. Bottom: Alpha-shapes of the projected point clouds for specific values of α .

When computing the RAUC index on the bidimensional projections, a value of 0.18 was obtained for PCA and 0.094 for UMAP. These results indicate, as expected, that UMAP generates a more well-defined geometric structure in the bidimensional space compared to PCA, which is clearly reflected in the behavior of the empty-space loss function.

In particular, UMAP preserves the alpha-shape consistently up to values close to 0.04. Beyond this point, increasing the radius of the alpha-shape begins to connect distant points, covering previously empty regions and leading to a progressive loss of the underlying geometric structure. In contrast, for PCA the space-filling process occurs more rapidly: small increases in α produce abrupt changes in the alpha-shape, resulting in an early loss of the optimal geometric structure, around 0.1.

From this perspective, the RAUC value suggests that the projection obtained via UMAP exhibits a geometric pattern closer to a clustering structure. In contrast, the projection obtained with PCA still shows some degree of clustering; however, it is closer to a random-like behavior when compared to UMAP. This is consistent with the visual representation shown in Figure 11, where the points in the UMAP projection appear organized into more compact and well-defined clusters, whereas PCA shows a more dispersed distribution with lower intra-cluster cohesion.

V. RESULTS AND DISCUSSION

Regression:

The experiments conducted on both the Boston Housing and simulated datasets demonstrate that the geometric structure

of predictor–response projections, as quantified by the RAUC index, provides meaningful insights into variable relevance, although it does not always directly translate into superior predictive performance.

In the Boston Housing dataset, variables such as `crim`, `rm`, and `lstat` exhibit the strongest geometric patterns according to RAUC, consistent with scatter plots showing compact, non-random distributions. However, the best-performing regression models in terms of RMSE often include additional variables with weaker individual geometric structure, highlighting that predictive accuracy depends not only on individual variable relevance but also on joint effects and interactions among predictors.

Similarly, in the simulated dataset, predictors with minimal individual geometric structure, such as X_5 , can still contribute to improved model performance when considered jointly with other variables. Conversely, predictors displaying strong geometric patterns, such as X_1 , X_2 , and X_4 , are consistently included in the best-performing models. This confirms that RAUC effectively identifies variables with meaningful structural relevance.

Overall, these results suggest that RAUC can guide the selection of predictors that exhibit coherent geometric organization with respect to the response, complementing traditional regression metrics and providing an additional layer of insight into the underlying structure of predictor–response relationships.

Dimensionality Reduction:

The analysis of bidimensional embeddings demonstrates the practical utility of the RAUC index. In Experiment 3, RAUC highlighted differences among UMAP projections, revealing configurations that produce more coherent clusters and better-preserved geometric structures compared to less optimal embeddings. This indicates that RAUC provides a quantitative assessment of spatial organization that goes beyond visual inspection.

Experiment 4 further emphasizes the role of RAUC in guiding hyperparameter selection for UMAP embeddings. By comparing RAUC values across various configurations, it is possible to identify embeddings that maintain well-defined, dense point patterns, effectively distinguishing projections that preserve the intrinsic geometry from those approaching a random spatial distribution.

Experiment 5 compared UMAP and PCA projections, demonstrating that UMAP consistently generates clearer clustering structures and more stable geometric patterns. Collectively, these results confirm that RAUC is a robust tool for evaluating low-dimensional embeddings, enabling the identification of projections with meaningful clusters and supporting the adjustment of hyperparameters to achieve optimal two-dimensional representations.

VI. CONCLUSIONS AND FUTURE WORK

This work demonstrates that the analysis of the geometric structure of predictor–response projections, through empty-space patterns and the RAUC index, provides complemen-

tary insights beyond classical correlation-based or linear approaches. By explicitly modeling the spatial organization induced by variable interactions, the proposed framework captures relevant structural information even in complex and nonlinear settings.

The results obtained from both simulated and real-world datasets show that RAUC effectively identifies predictors associated with persistent geometric structures, typically reflected in compact and non-random point clouds. At the same time, the experiments highlight that predictive performance is not solely determined by individual geometric relevance, but also by joint effects and interactions among variables. In this sense, RAUC should be understood as a complementary criterion that provides a principled way to assess the geometric coherence of predictor–response relationships.

In the context of dimensionality reduction, the experiments demonstrate that RAUC serves as a robust quantitative measure for evaluating the geometric quality of low-dimensional embeddings. In particular, it enables the identification of projections that preserve clustering structures and deviate from Complete Spatial Randomness, while also providing a systematic tool for hyperparameter selection in methods such as UMAP. This highlights its potential as an objective alternative to purely visual assessment.

Future research will focus on extending this framework in several directions. First, we aim to investigate its behavior across a broader range of dimensionality reduction techniques and datasets, with particular emphasis on the role of hyperparameters and sample size in shaping geometric structures. Second, further analysis of highly structured or atypical geometric patterns may provide deeper insights into nonlinear dependencies between variables. Third, the application of RAUC will be explored in additional machine learning tasks, including clustering, classification, and geometric extensions such as Riemannian principal component analysis. Finally, integrating RAUC into automated variable selection procedures represents a promising direction for developing geometry-aware learning pipelines.

Overall, the RAUC index provides a clear and intuitive way to understand how data are organized. This perspective opens new possibilities for analyzing patterns and relationships in a more accessible way, especially in cases where traditional methods fail to adequately describe how data are grouped or distributed.

VII. ACKNOWLEDGMENTS

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DATA AND CODE AVAILABILITY

All source code, generated datasets, and configuration settings used in this study are publicly available in the GitHub repository:

<https://github.com/DylanBC09/Geometric-Analysis-of-Point-Processes-with-Applications-in-Machine-Learning>.

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